

MA 301

Test 2

Fall - 2014

Problem	points	out of
1		12
2		5
3		5
4		5
5		5
Score		32

Name: SOLUTIONS

HR: _____

Signature: _____

Problem 1: Let $z_1 = -3i$, $z_2 = 3 - 3i$, $z_3 = -5$, $z_4 = -\sqrt{3} - i$.

(a) Perform the following operations and leave answer in standard form. (No Decimals)

$$\triangleright z_1 - z_4 = -3i + \sqrt{3} + i = \boxed{\sqrt{3} - 2i}$$

$$\triangleright \frac{z_4}{z_2} = \frac{-\sqrt{3} - i}{3 - 3i} = \frac{(-\sqrt{3} - i)(3 + 3i)}{9 + 9} = \frac{-3\sqrt{3} - 3\sqrt{3}i - 3i + 3}{18} = \frac{3 - 3\sqrt{3}}{18} + \frac{-3\sqrt{3}i - 3i}{18}$$

$$\text{or } \boxed{\frac{1 - \sqrt{3}}{6} - \frac{1 + \sqrt{3}}{6}i}$$

(b) Write z_1, z_2, z_3, z_4 in Euler's polar form. (No Decimals)

$$r = 3$$

$$\theta = -\frac{\pi}{2}$$

$$\triangleright z_1 = 3e^{-\frac{\pi}{2}i}$$

$$r = \sqrt{18}$$

$$\theta = -\frac{\pi}{4}$$

$$\triangleright z_2 = 3\sqrt{2}e^{-\frac{\pi}{4}i}$$

$$r = 5$$

$$\theta = \pi$$

$$\triangleright z_3 = 5e^{\pi i}$$

$$r = 2$$

$$\theta = \frac{7\pi}{6}$$

$$\triangleright z_4 = 2e^{\frac{7\pi}{6}i}$$

$$\theta = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}$$

(c) Perform the following operations and leave answer in Euler's polar form. (No Decimals)

$$\triangleright z_1 z_2 = (3e^{-\frac{\pi}{2}i})(3\sqrt{2}e^{-\frac{\pi}{4}i}) = 9\sqrt{2}e^{-\frac{3\pi}{4}i}$$

$$\triangleright (z_1 z_2)^6 = (9\sqrt{2}e^{-\frac{3\pi}{4}i})^6 = 9^6 \cdot 2^3 e^{-\frac{9\pi}{2}i}$$

(d) Solve the complex equations for x and leave your solutions in Euler's Form: $re^{i\theta}$. (No Decimals)

$$\triangleright \frac{x^{1/6}}{z_1} - z_2 = 0 \Rightarrow x^{1/6} = z_1 z_2 \Rightarrow x = (z_1 z_2)^6 = 4251528e^{-\frac{9\pi}{2}i}$$

$$\triangleright x^3 - z_3 = 0 \Rightarrow x = z_3^{1/3} = (5e^{\pi i})^{1/3} = 5^{1/3} \cdot e^{\frac{\pi + 2\pi k}{3}i} \quad k = 0, 1, 2$$

$$= 5^{1/3} e^{\frac{\pi}{3}i} \quad k = 0$$

$$= 5^{1/3} e^{\pi i} \quad k = 1$$

$$= 5^{1/3} e^{\frac{5\pi}{3}i} \quad k = 2$$

2: Let $F = [-3xz^2, 6, 5x^2y]$ and S is the parabolic cylinder $S: y = x^2$ where $0 \leq x \leq 2$ and $0 \leq z \leq 3$. Normal vector point away from the y -axis. Compute the flux of F through the parabolic cylinder S

let $u = x, v = z$

$$\vec{r}(u, v) = [u, u^2, v]$$

$$\vec{r}_u = [1, 2u, 0]$$

$$\vec{r}_v = [0, 0, 1]$$

$$\vec{N} = \vec{r}_u \times \vec{r}_v = [2u, -1, 0]$$

$$\vec{F}(\vec{r}) = [-3uv^2, 6, 5u^2]$$

$$\vec{F}(\vec{r}) \cdot \vec{N} = -6u^2v^2 - 6$$

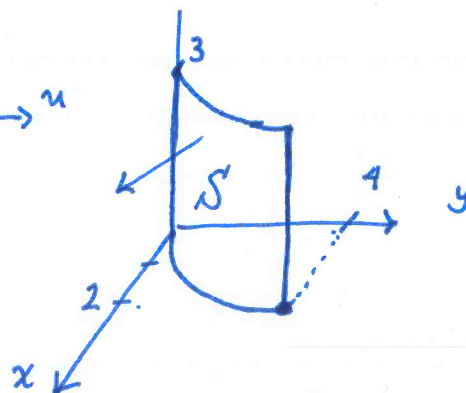
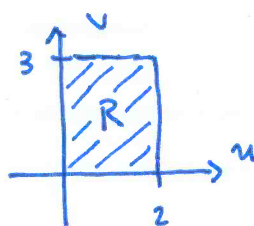
$$\text{Flux} = \int_0^2 \int_0^3 -6u^2v^2 - 6 \, dv \, du$$

$$= \int_0^2 -2u^2v^3 - 6v \Big|_0^3 \, du = \int_0^2 -54u^2 - 18 \, du$$

$$= -18u^3 - 18u \Big|_0^2$$

$$= -18(8) - 18(2)$$

$$= \boxed{-180}$$



check: $\vec{r}(2, 0) = [2, 4, 0]$

$$\vec{N}(2, 0) = [4, -1, 0]$$

Correct orientation.

Problem 3: Let $\vec{F}$ and S be the same as in **Problem 2**. If C be the boundary of S . Compute $\oint_C \vec{F} \cdot d\vec{r}$

$$\vec{F} = [-3xz^2, 6, 5x^2y]$$

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S \text{curl } \vec{F} \cdot \vec{n} \, dS = (*)$$

$$\text{curl } \vec{F} = [5x^2 - 0, -6xz - 10xy, 0 - 0] = [5x^2, -6xz - 10xy, 0]$$

Use the same $\vec{r}(u, v)$ as in **Problem 2**:

$$\vec{r}(u, v) = [u, u^2, v] \Rightarrow \vec{N} = [2u, -1, 0]$$

$$\text{curl } \vec{F}(\vec{r}) = [5u^2, -6uv - 10u^3, 0]$$

$$\text{curl } \vec{F}(\vec{r}) \cdot \vec{N} = 10u^3 + 6uv + 10u^3 = 20u^3 + 6uv$$

$$\begin{aligned} (*) &= \int_0^2 \int_0^3 (20u^3 + 6uv) \, dv \, du = \int_0^2 \left(20u^3 v + 3uv^2 \right) \Big|_0^3 \, du \\ &= \int_0^2 (60u^3 + 27u) \, du \\ &= \left(15u^4 + \frac{27}{2}u^2 \right) \Big|_0^2 \\ &= 15(16) + 27(2) \\ &= 294 \end{aligned}$$

Problem 4: Let $\mathbf{F} = [x, ye^z, e^z]$ and S the surface (outward orientation) of the box $|x| \leq 1, |y| \leq 2, -1 \leq z \leq 7$. Compute the flux through S .

$$\operatorname{div} \vec{F} = 1 + e^z + e^z = 1 + 2e^z$$

closed surface

$$\begin{aligned} \text{Flux} &= \iint_S \vec{F} \cdot \vec{n} \, dS = \iiint_T \operatorname{div} \mathbf{F} \, dV \\ &= \int_{-1}^7 \int_{-2}^2 \int_{-1}^1 (1 + 2e^z) \, dx \, dy \, dz \\ &= \int_{-1}^7 \int_{-2}^2 (1 + 2e^z)(2) \, dy \, dz \\ &= 8 \int_{-1}^7 (1 + 2e^z) \, dz \\ &= 8 \left(z + 2e^z \right) \Big|_{-1}^7 \\ &= 8(7 + 2e^7 + 1 - 2e^{-1}) \\ &= 64 + 16e^7 - 16e^{-1} \\ &\approx 17604.24 \end{aligned}$$

Problem 5: Let $F = [x^3 - y^3, y^3 - z^3, z^3 - x^3]$ and S is the surface of the solid bounded by:

- (i) quarter sphere $x^2 + y^2 + z^2 = 25$ ($y \geq 0, z \geq 0$);
- (ii) the plane $z = 0$;
- (iii) the plane $y = 0$

with outward pointing normal. Compute the flux through S .

$$\operatorname{div} \vec{F} = 3x^2 + 3y^2 + 3z^2$$

$$\text{Flux} = \iint_S \vec{F} \cdot \vec{n} \, dS$$

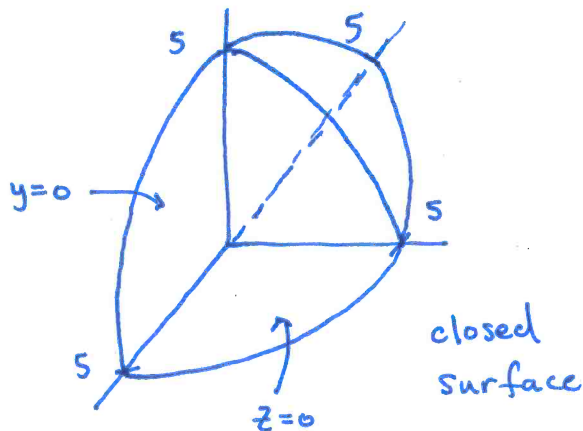
$$= \iiint_T \operatorname{div} F \, dV$$

$$= \iiint_T 3(x^2 + y^2 + z^2) \, dV$$

$$= \int_0^{\pi/2} \int_0^{\pi} \int_0^5 3\rho^2 \cdot \rho^2 \sin \phi \, d\rho \, d\theta \, d\phi$$

$$= 3\pi \left(\int_0^{\pi/2} \sin \phi \, d\phi \right) \left(\int_0^5 \rho^4 \, d\rho \right)$$

$$= 3\pi \left(-\cos \phi \right) \Big|_0^{\pi/2} \frac{\rho^5}{5} \Big|_0^5 = 3\pi (1)(5^4) = 1875\pi \approx 5887.5$$



$$0 \leq \rho \leq 5$$

$$0 \leq \theta \leq \pi$$

$$0 \leq \phi \leq \frac{\pi}{2}$$